**A** 24 held-out phantoms ($T_1$s uniform over the $T_1$-plate range, as in training)

| Method | Memory | Budget | Mean scan time | MAPE % [95% CI] | Scans < 5% | Worst scan MAPE % |
|---|---|---|---|---|---|---|
| **RL** | **TI histogram** | **560 s** | **541 s** | **2.93 [2.58, 3.30]** | **95.8%** | **5.13** |
| RL | none | 560 s | 548 s | 4.16 [3.41, 4.99] | 66.7% | 9.50 |
| RL | LSTM† | 560 s | 525 s | 7.31 [6.10, 8.65] | 25.0% | 15.02 |
| Fixed, distilled from RL policy§ | – | 560 s | 549 s | 3.34 [2.30, 4.86] | 91.7% | 17.87 |
| Fixed log TI grid | – | 560 s | 512 s | 5.12 [4.52, 5.71] | 45.8% | 8.67 |
| Cramér-Rao optimal | – | 560 s | 558 s | 7.74 [7.08, 8.42] | 8.3% | 11.72 |
| RL | none | 240 s | 227 s | 4.62 [4.06, 5.19] | 62.5% | 7.74 |

**B** 96 clustered-$T_1$ phantoms ($T_1$s within ±15% of a random centre; policy not retrained)

| Method | Memory | Budget | Mean scan time | MAPE % [95% CI] | Scans < 5% | Worst scan MAPE % |
|---|---|---|---|---|---|---|
| **RL** | **TI histogram** | **560 s** | **545 s** | **2.76 [2.58, 2.95]** | **96.9%** | **5.35** |
| Fixed, distilled from RL policy§ | – | 560 s | 549 s | 12.20 [8.02, 16.94] | 78.1% | 91.47 |
| Fixed log TI grid | – | 560 s | 512 s | 8.77 [8.02, 9.54] | 18.8% | 21.61 |
| Cramér-Rao optimal | – | 560 s | 558 s | 8.53 [7.95, 9.16] | 10.4% | 18.56 |

† Undertrained: the LSTM ran 4.6× slower, so within the same 9 h budget it took 7.3× fewer full-Bloch steps. Its best checkpoint predates full-Bloch training, so this is not a fair test of recurrent memory.

§ One fixed schedule: per-block median TI and TR of the TI-histogram policy's schedules on 24 validation scans (13 blocks, 549 s).